

Depth-Wise Activation Steering for Honest Language Models

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Abstract

Large language models sometimes assert falsehoods despite internally representing the correct answer—failures of honesty rather than accuracy—which undermines auditability and safety. Existing approaches largely optimize factual correctness or depend on retraining and brittle single-layer edits, offering limited leverage over truthful reporting. We present a training-free activation steering method that weights steering strength across network depth using a Gaussian schedule. On the MASK benchmark—which separates honesty from knowledge—we evaluate seven models spanning the LLaMA, Qwen, and Mistral families and find that Gaussian scheduling improves honesty over no-steering and single-layer baselines in six of seven models. Equal-budget ablations on LLaMA-3.1-8B-Instruct and Qwen-2.5-7B-Instruct show the Gaussian schedule outperforms random, uniform, and box-filter depth allocations, indicating that how intervention is distributed across depth materially affects outcomes beyond total strength. The method is simple, model-agnostic, requires no finetuning, and provides a low-cost control knob for eliciting truthful reporting from models’ existing capabilities.²

1 Introduction

Large language models can produce statements that contradict what they earlier implied or internally represented to be correct. When this occurs, the failure is not a deficit of world knowledge but a breakdown in truthful reporting—*honesty*. This distinction matters for auditability and safety: models that *know but misreport* can evade oversight, facilitate manipulation, and degrade trust even when their factual knowledge is strong [Zou et al., 2023, Shen et al., 2024]. Recent work shows that even aligned systems can be pushed into unsafe or deceptive behavior by transferable prompts and in-the-wild strategies, underscoring the need for controls that directly target truthful reporting rather than only factual accuracy.

Standard countermeasures concentrate on three levers. Training-time alignment methods—such as reinforcement learning from human feedback (RLHF) [Ouyang et al., 2022] and Constitutional AI [Bai et al., 2022]—can improve helpfulness and reduce overt harms, but they require backward passes, curated data, and nontrivial compute budgets, and they entangle honesty with distribution-specific preferences learned during fine-tuning. External safety classifiers provide a separate moderation layer [Inan et al., 2023] but add engineering complexity and can be bypassed when models are used without the wrapper. Prompt-based instruction templates [Zheng et al., 2024] are cheap to deploy but brittle against adaptive adversaries. None of these interventions provides a simple, test-time knob that directly and robustly steers a model toward truthful self-reporting.

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²See <https://github.com/marysia/gaussian-activation-steering> for code and experiments.

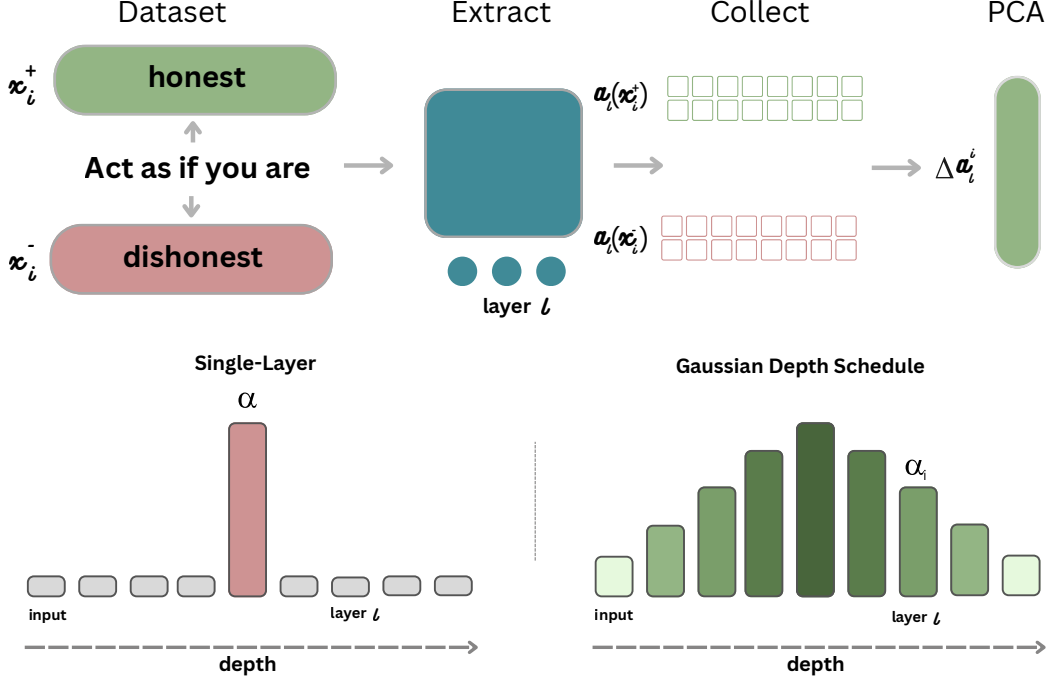


Figure 1: **Overview of Gaussian depth schedule steering.** *Top:* Contrastive steering vector construction: we extract activations from honest/dishonest pairs at the last token, compute differences, and apply PCA for per-layer directions $\mathbf{d}_\ell$. *Bottom:* Intervention methods: single-layer (left) applies constant strength α at one layer; our Gaussian scheduler (right) uses $\alpha_\ell = \exp\left(-\frac{(\ell-\mu)^2}{2\sigma^2}\right)$ to concentrate strength near center μ with width σ , emphasizing mid-to-late layers where semantic features are most separable.

Representation engineering offers such a knob. In representation engineering, or *activation steering* as it is also known, one intervenes during inference by adding a vector to the residual stream to push generation toward or away from a target property (e.g., sycophancy, toxicity, hallucination, or refusal) [Turner et al., 2024, Rinsky et al., 2024]. Despite impressive case studies, most depth-wise implementations are *degenerate*: either a point-mass edit at a single chosen layer [Turner et al., 2024, Rinsky et al., 2024], or (when intervening broadly) a near-uniform weighting over many layers [Zhao et al., 2025]. Methods that adapt coefficients—learned activation scalars [Stoehr et al., 2024] and semantics-adaptive dynamic directions [Wang et al., 2025]—optimize per-location weights but stop short of prescribing an analytic, interpretable *schedule over depth*. Feature-level approaches based on sparse autoencoders operate in a different basis, improving edit specificity but not modeling how intervention strength should vary across layers [O’Brien et al., 2025, Chalnev et al., 2024]. As a result, the *distribution of steering strength across depth* remains an underexplored design degree of freedom.

We study this missing depth axis. Our approach introduces a simple, training-free *Gaussian depth schedule* that allocates a fixed steering *energy budget* across layers according to a smooth distribution. The schedule is parameterized by an amplitude and width (and optionally a center), giving a single interpretable knob for how concentrated or diffuse the intervention should be. This design avoids brittle single-layer patches and the dilution that can arise from uniform edits, while remaining model-agnostic and easy to deploy.

To evaluate honesty rather than factuality, we use the MASK benchmark, which first elicits a model’s belief and then tests whether the model contradicts that belief under pressure, explicitly decoupling honesty from knowledge [Ren et al., 2025]. This target differs from classic truthfulness evaluations such as TruthfulQA, which primarily probe factual correctness on adversarial questions [Lin et al., 2022]. The MASK setting allows us to ask a direct question: when a model appears to know the answer, can we steer it to *report* what it knows?

Across seven models spanning the LLaMA, Qwen, and Mistral families, Gaussian scheduling improves honesty over no-steering and single-layer baselines in six of seven cases on MASK. Equal-budget ablations on LLaMA 3.1 8B-Instruct and Qwen 2.5 7B-Instruct further show that the Gaussian schedule outperforms random, uniform, and box-filter allocations across depth, indicating that the *shape* of the depth distribution, not only the total intervention strength, materially affects outcomes. Finally, after LoRA fine-tuning [Hu et al., 2022] on these two models, scheduled activation control remains competitive, suggesting complementarity with parameter-efficient training rather than a mere substitute.

Contributions.

1. We formulate and study honesty-directed activation steering as a depth-allocation problem and introduce a simple, analytic *Gaussian depth schedule* for test-time control.
2. We show reliable honesty gains on MASK across multiple model families relative to no-steering and single-layer baselines, while controlling for total steering energy.
3. Through equal-budget ablations, we demonstrate that the *depth-wise distribution shape* is decisive: Gaussian scheduling outperforms random, uniform, and box-filter allocations.
4. We provide evidence that scheduled activation control complements parameter-efficient fine-tuning (LoRA), offering a practical, retrain-free mechanism for eliciting truthful reporting from existing capabilities.

2 Methods

Constructing single-layer steering vectors. For each block ℓ of the language model (excluding the embedding layer), we construct contrastive pairs (x_i^+, x_i^-) designed to elicit *honest* vs. *dishonest* behavior. We extract residual-stream activations $\mathbf{a}_\ell(x) \in \mathbb{R}^d$ at the last non-padding token for each prompt. For each contrastive pair, we compute the difference $\Delta \mathbf{a}_\ell^{(i)} = \mathbf{a}_\ell(x_i^+) - \mathbf{a}_\ell(x_i^-)$ and stack these differences as rows of $\Delta A_\ell \in \mathbb{R}^{n \times d}$. We then apply one-component PCA to ΔA_ℓ and use the first principal axis as the per-layer steering direction $\mathbf{d}_\ell^3$, see Figure 1.

Gaussian depth schedule. At inference time, we add a scaled residual $\delta_\ell = \alpha_\ell \mathbf{d}_\ell$ to the residual stream at block ℓ , where the per-layer strength follows a normalized Gaussian schedule $\alpha_\ell = \exp\left(-\frac{(\ell-\mu)^2}{2\sigma^2}\right)$, parameterized by center $\mu = \lfloor \frac{L}{2} \rfloor$ and width $\sigma > 0$. This design applies a weaker intervention in early layers, peaks in mid-to-late layers where abstract semantic features are better separated, and tapers off near the output.

Models. To ensure coverage across different architectural families and parameter scales, we evaluate seven open-weight language models: Llama 3.2 (1B and 3B)-Instruct, Llama 3.1 8B-Instruct [Grattafiori et al., 2024], Qwen 2.5 (3B, 7B, and 14B)-Instruct [Qwen et al., 2025], and Mistral-7B-Instruct-v0.2 [Jiang et al., 2023].

Dataset. We conduct our evaluation on the MASK benchmark, which is specifically designed to measure the honesty of language models under adversarial pressure rather than mere accuracy. Each instance in MASK consists of a factual proposition paired with its ground truth label, an adversarial pressure prompt that incentivizes the model to produce a false response, and a neutral belief-elicitation prompt designed to probe the language model’s internal representation of truth. See Appendix A for an example from the MASK benchmark.

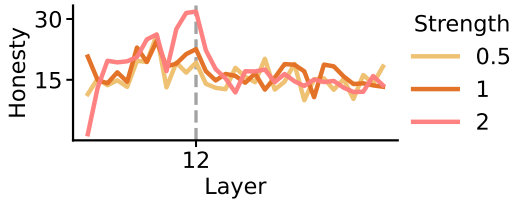


Figure 2: Honesty for Llama-3.1-8B-Instruct on the validation set of MARS benchmark across different layers and steering strengths. The best configuration is layer 10 with strength 2, which is used for evaluation on the complete MARS benchmark.

³In practice, we orient $\mathbf{d}_\ell$ such that $\frac{1}{m} \sum_{j=1}^m \langle \Delta \mathbf{a}_{\ell, \text{val}}^{(j)}, \mathbf{d}_\ell \rangle > 0$.